

Problem

The voltage across a $1\text{-}\Omega$ resistor is $v(t) = te^{-2t}u(t)$ V.

(a) What is the total energy absorbed by the resistor?

(b) What fraction of this energy absorbed is in the frequency band $-2 \leq \omega \leq 2$?

Step-by-step solution

Step 1 of 5

Consider the voltage across a $1\text{ }\Omega$ resistor is $v(t) = te^{-2t}u(t)$ V.

Calculate the total energy absorbed by the resistor using the following formula.

$$\begin{aligned} W &= \int_{-\infty}^{\infty} v^2(t) dt \\ &= \int_0^{\infty} (te^{-2t})^2 dt \\ &= \int_0^{\infty} t^2 e^{-4t} dt \\ &= \int_0^{\infty} \left(t^2 \frac{d}{dt} \left(\frac{e^{-4t}}{-4} \right) \right) dt \end{aligned}$$

Apply the integral by parts formula, $\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$, to determine W .

$$\begin{aligned} W &= \int_0^{\infty} \left(t^2 \frac{d}{dt} \left(\frac{e^{-4t}}{-4} \right) \right) dt \\ &= t^2 \left(\frac{e^{-4t}}{-4} \right) \Big|_{t=0}^{\infty} - \int_0^{\infty} \left(\left(\frac{e^{-4t}}{-4} \right) \frac{d}{dt} (t^2) \right) dt \\ &= t^2 \left(\frac{e^{-4t}}{-4} \right) \Big|_{t=0}^{\infty} - \int_0^{\infty} \left(\left(\frac{e^{-4t}}{-4} \right) (2t) \right) dt \\ &= 0 - \int_0^{\infty} (2t) \frac{d}{dt} \left(\frac{e^{-4t}}{-4(-4)} \right) dt \end{aligned}$$

Step 2 of 5

Apply the integral by parts formula, $\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$, to determine W .

$$\begin{aligned}
 W &= 0 - \int_0^{\infty} \left((2t) \frac{d}{dt} \left(\frac{e^{-4t}}{-4(-4)} \right) \right) dt \\
 &= - \left(\frac{(2t)e^{-4t}}{-4(-4)} \right) \Bigg|_{t=0}^{\infty} - \int_0^{\infty} \left(\frac{e^{-4t}}{-4(-4)} \right) \left(\frac{d}{dt}(2t) \right) dt \\
 &= - \left(0 - \int_0^{\infty} 2 \left(\frac{e^{-4t}}{-4(-4)} \right) dt \right) \\
 &= \left(2 \int_0^{\infty} \left(\frac{e^{-4t}}{-4(-4)} \right) dt \right) \\
 &= 2 \left(\frac{1}{16} \right) \left(\frac{-1}{4} \right) e^{-4t} \Bigg|_{t=0}^{\infty} \\
 &= 2 \left(\frac{1}{16} \right) \left(\frac{-1}{4} \right) (0 - 1) \\
 &= 0.0313 \text{ J}
 \end{aligned}$$

Thus, the total energy observed by the resistor is 0.0313 J.

Step 3 of 5

Calculate the $v(t)$ in frequency domain.

$$V(\omega) = \mathcal{F}[te^{-2t}u(t)]$$

Write the Fourier transform pairs that are used to find $V(\omega)$.

$$te^{-at} \xleftrightarrow{\mathcal{F}} \frac{1}{(j\omega + a)^2}$$

Substitute 2 in place of a to find $V(\omega)$.

$$V(\omega) = \frac{1}{(j\omega + 2)^2}$$

Step 4 of 5

Calculation of $|V(\omega)|^2$:

$$\begin{aligned} |V(\omega)|^2 &= V(\omega)V^*(\omega) \\ &= \left(\frac{1}{(j\omega+2)^2} \right) \left(\frac{1}{(-j\omega+2)^2} \right) \\ &= \frac{1}{(4+\omega^2)^2} \end{aligned}$$

Calculate the energy observed by the resistor in frequency band $-2 \leq \omega \leq 2$.

$$\begin{aligned} W_0 &= \frac{1}{2\pi} \int_{-2}^2 |V(\omega)|^2 d\omega \\ &= \frac{1}{2\pi} \int_{-2}^2 \left(\frac{1}{(4+\omega^2)^2} \right) d\omega \\ &= \frac{2}{2\pi} \int_0^2 \left(\frac{1}{(4+\omega^2)^2} \right) d\omega \\ &= \frac{1}{\pi} \frac{1}{(2(4))} \left(\frac{\omega}{\omega^2+4} + \left(\frac{1}{2} \right) \tan^{-1} \left(\frac{\omega}{2} \right) \right) \Bigg|_{\omega=0}^2 \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} W_0 &= \frac{1}{8\pi} \left(\left(\frac{2}{4+2^2} + \frac{1}{2} \tan^{-1} \left(\frac{2}{2} \right) \right) - \left(0 + \frac{1}{2} \tan^{-1} \left(\frac{0}{2} \right) \right) \right) \\ &= \frac{1}{8\pi} \left(\left(\frac{2}{8} + \frac{1}{2} \left(\frac{\pi}{4} \right) \right) - (0+0) \right) \\ &= \frac{1}{32\pi} + \frac{1}{64} \end{aligned}$$

Step 5 of 5

Calculate the fraction of energy absorbed by the frequency band $-2 \leq \omega \leq 2$ in the total energy.

$$\begin{aligned} \frac{W_0}{W} &= \frac{\left(\frac{1}{32\pi} + \frac{1}{64} \right)}{0.0313} \\ &= \frac{(9.947 \times 10^{-3}) + 0.0156}{0.0313} \\ &= \frac{0.0256}{0.0313} \\ &= 81.79\% \end{aligned}$$

Thus, the fraction of energy absorbed by the frequency band $-2 \leq \omega \leq 2$ in the total energy is

81.79%.